

100 DAYS OF CODE – DAY 4

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Problem Statement

Correct the given C program, algorithm, and flowchart to perform multiplication of two integers without using the multiplication (*) operator. Implement the solution using repeated addition and ensure that it correctly handles positive numbers, negative numbers, and zero.

Objective

To implement multiplication of two integers using repeated addition instead of the multiplication operator and verify the program using different test cases.

Given Algorithm

1. Define num1, num2, product and index as integer variables.
2. Read the values of num1 and num2.
3. Initialize product = 0 and index = 1.
4. Repeat until index is greater than num2:
5. product = product + num1
6. index = index + 1
7. Display the product.
8. Stop.

Mistakes in the Given Algorithm

- It works only for positive values of num2.
- It does not handle negative numbers.
- It does not correctly handle multiplication when one input is zero.
- It does not determine the sign of the final result.

Correct Algorithm

9. Start.
10. Read num1 and num2.
11. Initialize product = 0.
12. Check whether the result should be negative.
13. Convert both numbers to their absolute values.
14. Repeat |num2| times and add num1 to product.
15. If required, change the sign of product.
16. Print the product.
17. Stop.

Conclusion

This experiment demonstrates multiplication using repeated addition instead of the '*' operator. The corrected algorithm successfully handles positive numbers, negative numbers, and zero, producing the expected result.